

Differential Equations and Slope Fields



Verifying solutions to differential equations

- 1 Verify that $y = Ce^{3x}$ is a solution to the differential equation $\frac{dy}{dx} = 3y$ for any constant C .
 - 2 Verify that $y = x^2 + C$ is a general solution to $\frac{dy}{dx} = 2x$, and find the particular solution satisfying $y(1) = 5$.
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Solving separable differential equations

- 3 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.
 - 4 Solve $\frac{dy}{dx} = 2xy$ given $y(0) = 1$.
 - 5 Solve $\frac{dy}{dx} = \frac{\cos x}{y^2}$ given $y(0) = 2$.
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Slope fields

- 6 For the differential equation $\frac{dy}{dx} = x - y$, find the slope at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, as would be drawn on a slope field.
 - 7 Explain how a slope field for $\frac{dy}{dx} = ky$ (exponential growth/decay) would look different for $k > 0$ versus $k < 0$.
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Exponential growth and decay models

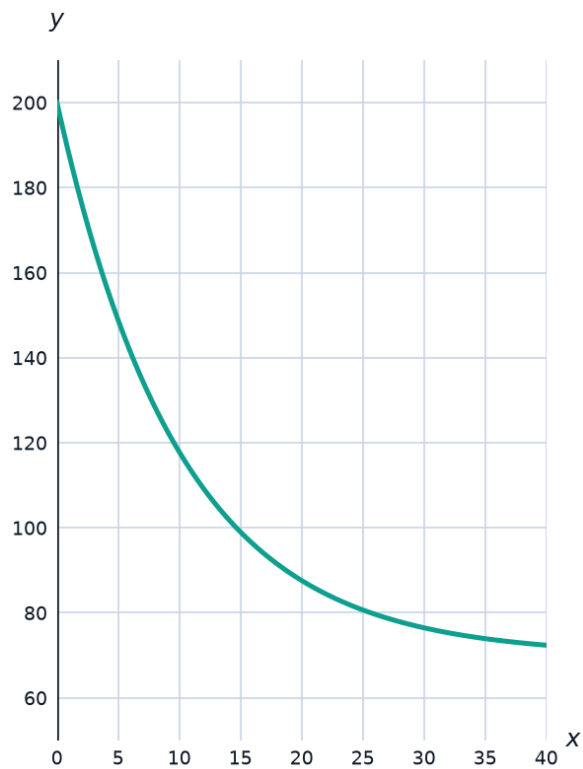
- 8 A bacteria culture grows according to $\frac{dP}{dt} = 0.2P$, with $P(0) = 500$. Find $P(t)$ and the population after 10 hours.
 - 9 A radioactive substance decays according to $\frac{dA}{dt} = -0.05A$. If $A(0) = 100$ grams, find how long it takes to decay to 50 grams (the half-life).
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Matching slope fields to differential equations

- 10 A slope field shows horizontal segments (slope 0) along the entire x -axis and along the entire y -axis, with slopes elsewhere resembling the product xy . Identify the differential equation this slope field represents.
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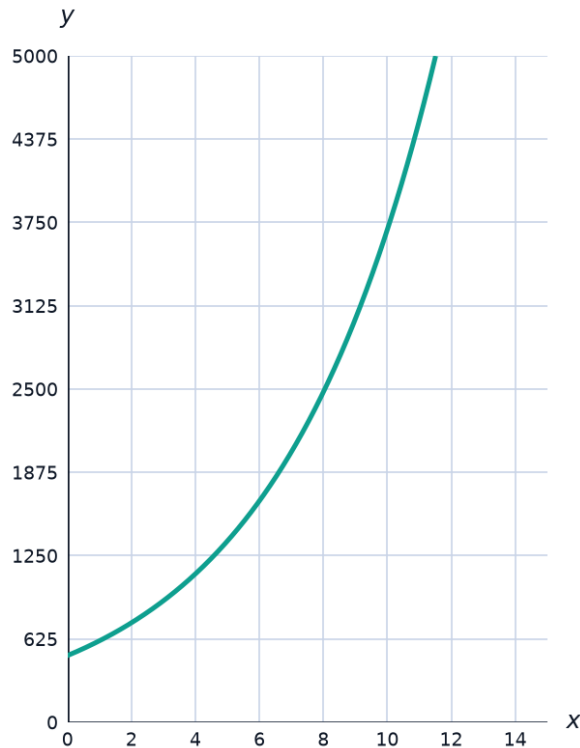
Newton's Law of Cooling

- 11 An object cools according to $\frac{dT}{dt} = -k(T - 70)$, where 70°F is room temperature. If $T(0) = 200^\circ\text{F}$ and $k = 0.1$, find $T(t)$.



Visualizing an exponential growth solution

- 12 The graph shows the bacteria population $P(t) = 500e^{0.2t}$ from the earlier problem. Use the graph to estimate how long it takes for the population to double from its initial value of 500, then verify algebraically using $\ln 2$.



A logistic-style differential equation

- 13 A population satisfies $\frac{dP}{dt} = 0.1P(1 - \frac{P}{200})$. Find the equilibrium solutions (where $\frac{dP}{dt} = 0$), and explain which one is stable for a population starting near $P = 10$.

Additional separable equations

- 14 Solve $\frac{dy}{dx} = \frac{y}{x}$ given $y(1) = 6$.
- 15 Solve $\frac{dy}{dx} = \frac{1}{y}$ given $y(0) = 4$.

A mixing problem modeled by a differential equation

- 16 A tank contains 50 L of pure water. Brine with concentration 3 kg/L flows in at 2 L/min, and the well-mixed solution drains at the same rate. If $y(t)$ is the amount of salt (kg) at time t , the differential equation is $\frac{dy}{dt} = 6 - \frac{2y}{50} = 6 - \frac{y}{25}$. Find the equilibrium value of y , and explain what it represents physically.

Multiple choice — easy 15

- 17 For the differential equation $\frac{dy}{dx} = -3x + 5y$, find the slope at the point $(-1, -4)$, as would be drawn on a slope field.
- a. -19
 - b. -17
 - c. -15
 - d. 23
- 18 For the differential equation $\frac{dy}{dx} = -3x - 5y$, find the slope at the point $(-2, 2)$, as would be drawn on a slope field.
- a. -2
 - b. -4
 - c. -6
 - d. 16
- 19 For the differential equation $\frac{dy}{dx} = -5x$, find the slope at the point $(0, -3)$, as would be drawn on a slope field.
- a. -2
 - b. 1
 - c. 0
 - d. 2

- 20 For the differential equation $\frac{dy}{dx} = 4x - 2y$, find the slope at the point $(-4, 4)$, as would be drawn on a slope field.
- a. -22
 - b. -26
 - c. -24
 - d. -8
- 21 For the differential equation $\frac{dy}{dx} = -x - 3y$, find the slope at the point $(3, 3)$, as would be drawn on a slope field.
- a. 6
 - b. -14
 - c. -10
 - d. -12
- 22 For the differential equation $\frac{dy}{dx} = 3x - 5y$, find the slope at the point $(3, -2)$, as would be drawn on a slope field.
- a. 17
 - b. 19
 - c. -1
 - d. 21
- 23 For the differential equation $\frac{dy}{dx} = -3x + 5y$, find the slope at the point $(3, 1)$, as would be drawn on a slope field.
- a. -4
 - b. -14
 - c. -6
 - d. -2

- 24 For the differential equation $\frac{dy}{dx} = 5x - y$, find the slope at the point $(-4, -3)$, as would be drawn on a slope field.
- a. -15
 - b. -19
 - c. -17
 - d. -23
- 25 For the differential equation $\frac{dy}{dx} = -5x + y$, find the slope at the point $(-2, -3)$, as would be drawn on a slope field.
- a. 13
 - b. 7
 - c. 9
 - d. 5
- 26 For the differential equation $\frac{dy}{dx} = 5x + y$, find the slope at the point $(-4, -4)$, as would be drawn on a slope field.
- a. -24
 - b. -16
 - c. -26
 - d. -22
- 27 For the differential equation $\frac{dy}{dx} = 0x + 2y$, find the slope at the point $(4, -4)$, as would be drawn on a slope field.
- a. -10
 - b. 8
 - c. -8
 - d. -6

- 28 For the differential equation $\frac{dy}{dx} = -2x - 4y$, find the slope at the point $(0, -4)$, as would be drawn on a slope field.
- a. 18
 - b. -16
 - c. 16
 - d. 14
- 29 For the differential equation $\frac{dy}{dx} = -3x + 2y$, find the slope at the point $(-4, 4)$, as would be drawn on a slope field.
- a. 20
 - b. 18
 - c. 4
 - d. 22
- 30 For the differential equation $\frac{dy}{dx} = 0x + 2y$, find the slope at the point $(4, 4)$, as would be drawn on a slope field.
- a. 8
 - b. -8
 - c. 10
 - d. 6
- 31 For the differential equation $\frac{dy}{dx} = -x + 2y$, find the slope at the point $(-4, -2)$, as would be drawn on a slope field.
- a. 8
 - b. 0
 - c. -2
 - d. 2

Multiple choice — medium 15

- 32 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 21$. Find $y(20)$ (the positive root).
- a. 21
 - b. 29
 - c. 20
 - d. 28
- 33 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(-9)$ (the positive root).
- a. 12
 - b. 9
 - c. 14
 - d. 15
- 34 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 24$. Find $y(18)$ (the positive root).
- a. 30
 - b. 18
 - c. 29
 - d. 24
- 35 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 8$. Find $y(-15)$ (the positive root).
- a. 17
 - b. 16
 - c. 8
 - d. 15

- 36 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 40$. Find $y(-9)$ (the positive root).
- a. 40
 - b. 39
 - c. 41
 - d. 9
- 37 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 7$. Find $y(24)$ (the positive root).
- a. 7
 - b. 25
 - c. 27
 - d. 24
- 38 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(9)$ (the positive root).
- a. 9
 - b. 12
 - c. 15
 - d. 14
- 39 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 24$. Find $y(10)$ (the positive root).
- a. 10
 - b. 25
 - c. 26
 - d. 24

- 40 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 7$. Find $y(-24)$ (the positive root).
- a. 25
 - b. 28
 - c. 7
 - d. 24
- 41 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 21$. Find $y(-20)$ (the positive root).
- a. 21
 - b. 29
 - c. 28
 - d. 20
- 42 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 20$. Find $y(15)$ (the positive root).
- a. 24
 - b. 25
 - c. 20
 - d. 15
- 43 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(-5)$ (the positive root).
- a. 12
 - b. 13
 - c. 16
 - d. 5

- 44 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 4$. Find $y(3)$ (the positive root).
- a. 7
 - b. 5
 - c. 4
 - d. 3
- 45 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 24$. Find $y(-10)$ (the positive root).
- a. 24
 - b. 10
 - c. 25
 - d. 26
- 46 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 16$. Find $y(12)$ (the positive root).
- a. 12
 - b. 19
 - c. 16
 - d. 20

Multiple choice — hard 15

- 47 A quantity grows according to $\frac{dP}{dt} = \frac{3}{20}P$, with $P(0) = 326$. Find $P(5)$, rounded to the nearest whole number.
- a. 587
 - b. 594
 - c. 794
 - d. 690

- 48 A quantity grows according to $\frac{dP}{dt} = \frac{3}{20}P$, with $P(0) = 425$. Find $P(12)$, rounded to the nearest whole number.
- a. 2957
 - b. 2571
 - c. 2185
 - d. 2213
- 49 A quantity grows according to $\frac{dP}{dt} = \frac{3}{20}P$, with $P(0) = 220$. Find $P(11)$, rounded to the nearest whole number.
- a. 1146
 - b. 974
 - c. 1317
 - d. 986
- 50 A quantity grows according to $\frac{dP}{dt} = \frac{1}{4}P$, with $P(0) = 805$. Find $P(10)$, rounded to the nearest whole number.
- a. 9807
 - b. 8336
 - c. 7638
 - d. 11278
- 51 A quantity decays according to $\frac{dP}{dt} = -\frac{1}{4}P$, with $P(0) = 473$. Find $P(7)$, rounded to the nearest whole number.
- a. 106
 - b. 95
 - c. 70
 - d. 82

- 52 A substance decays according to $\frac{dA}{dt} = -\frac{1}{10}A$. If $A(0) = 126$ grams, find how long it takes to decay to 84 grams, to one decimal place.
- a. 9.1
 - b. 2.8
 - c. 4.1
 - d. 5.3
- 53 A substance decays according to $\frac{dA}{dt} = -\frac{2}{25}A$. If $A(0) = 120$ grams, find how long it takes to decay to 40 grams, to one decimal place.
- a. 9.6
 - b. 13.7
 - c. 18.7
 - d. 17.9
- 54 A substance decays according to $\frac{dA}{dt} = -\frac{1}{20}A$. If $A(0) = 267$ grams, find how long it takes to decay to 89 grams, to one decimal place.
- a. 22.0
 - b. 28.6
 - c. 27.0
 - d. 15.4
- 55 A substance decays according to $\frac{dA}{dt} = -\frac{1}{20}A$. If $A(0) = 162$ grams, find how long it takes to decay to 54 grams, to one decimal place.
- a. 27.0
 - b. 22.0
 - c. 28.6
 - d. 15.4

- 56 A substance decays according to $\frac{dA}{dt} = -\frac{3}{100}A$. If $A(0) = 64$ grams, find how long it takes to decay to 32 grams, to one decimal place.
- a. 30.0
 - b. 16.2
 - c. 23.1
 - d. 28.1
- 57 A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 9 - \frac{y}{25}$. Find the equilibrium value of y .
- a. 216
 - b. 34
 - c. 225
 - d. 250
- 58 A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 5 - \frac{y}{35}$. Find the equilibrium value of y .
- a. 170
 - b. 40
 - c. 175
 - d. 210
- 59 A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 10 - \frac{y}{30}$. Find the equilibrium value of y .
- a. 300
 - b. 290
 - c. 330
 - d. 40

- 60 A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 2 - \frac{y}{15}$. Find the equilibrium value of y .
- a. 30
 - b. 17
 - c. 45
 - d. 28
- 61 A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 5 - \frac{y}{57}$. Find the equilibrium value of y .
- a. 280
 - b. 62
 - c. 285
 - d. 342